

Visionary

12+3 · BEHAVIOR + follow-ups · NO Discord/Drive

BEHAVIOR winners = $\pi_{0.5}$ +FM+SigLIP. JEPA = parallel latent imagination. Hybrid JEPA+FM = PROPOSAL ONLY. Dense unlock = contact geometry / depth for household.

Shared spine — tattoo every deck

Six claims that must land in every seat

- 1 Unlock dense features in the JEPA world-model line — L_ctx + deep SS + VisionMix/ViT-G.
- 2 CLEAR lineage: I-JEPA → V-JEPA → V-JEPA 2 → V-JEPA 2.1 → 2-AC.
- 3 Franka grasp 60→80%; latent nav ~10× cheaper than NWM.
- 4 Triangle: SigLIP/DINO | MAE | JEPA WM (now dense).
- 5 Must: NYU 0.307; ADE 22.2→47.9; Ego4D 7.71; Tables 6-7; L_ctx alone hurts globals / deep SS restores.
- 6 SSv2 77.7; EK100 40.8; Tartan ATE 5.687 / 10.6s vs NWM 103.2s.

BEHAVIOR landscape — two parallel bets

Winners $\neq$ JEPA children

BEHAVIOR winners (2025)

$\pi_{0.5}$ + flow-matching action experts + SigLIP/PaliGemma
— generative short-horizon motor interface.
Language-aligned towers ship today.

JEPA parallel bet

V-JEPA 2 / 2.1 / 2-AC = latent predictive
imagination + CEM / diffusion planning with image
goals and almost no task demos.

Dense unlock matters for BEHAVIOR: contact geometry, object boundaries, depth for grasp affordances —
exactly what VJ2 lacked (NYU 0.307, VOC 85.0, tracking ~70).

Hybrid JEPA+FM — PROPOSAL ONLY

Not demonstrated in this paper · do not invent as result

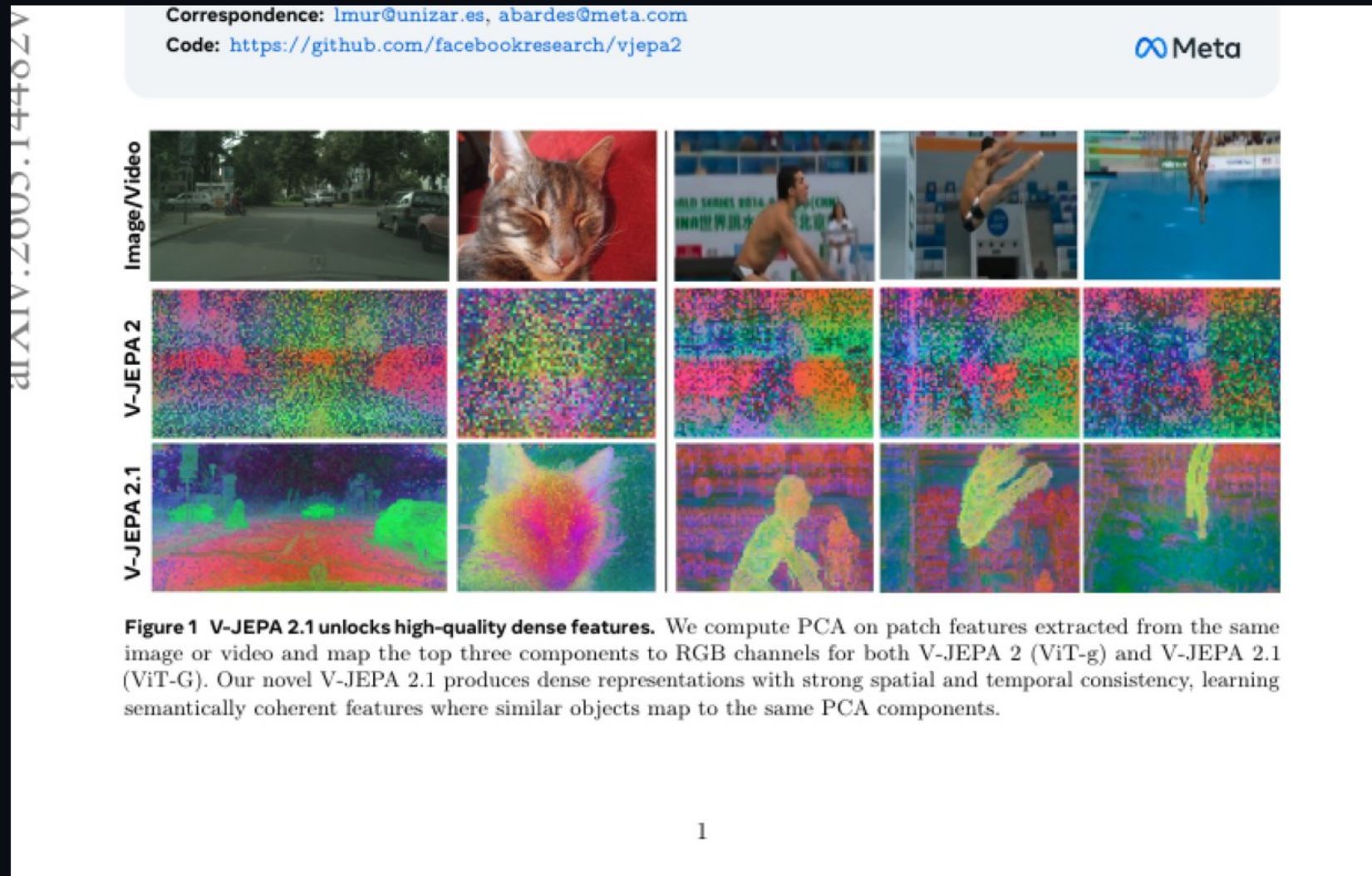
PROPOSAL (research agenda)

V-JEPA 2.1 latent rollout for long-horizon feasibility / goal imagination + FM action expert for motor execution. Long-horizon household = where pixel video WMs waste capacity and pure BC compounds error.

FORBIDDEN phrasing: 'V-JEPA 2.1 + $\pi_{0.5}$ hybrid wins BEHAVIOR.' ALLOWED: 'A natural next experiment is ...'

Dense unlock → contact / depth trustworthy

Fig. 1 — why household grasp imagines better



1

Spatially coherent features are necessary for grasp-along-depth and latent MPC; not a SigLIP substitute.

Follow-up research agenda

Six tickets — measure, don't slogan

- 1** AC on OXE / DROID / BEHAVIOR-1K with 2.1 encoder — sample efficiency vs π_0 BC.
- 2** Goal-image CEM inside BEHAVIOR scenes; success vs language-goal VLAs.
- 3** Dense aux heads (depth/seg/track) for WM planning — does grasp exceed 80%?
- 4** Cross-embodied latents (camera + proprio).
- 5** Uncertainty-aware predictors for contact-rich failure anticipation.
- 6** Scale 2B→7B + denser WM heads (paper future work; Karypidis/Bojanowski).

New apps — NO Discord / NO Drive

Beyond chat/docs demos

Zero-shot rearrange

Image goals on new kitchens — 2-AC recipe + denser encoder.

Fast latent nav

Indoor robots — 10x vs VAE latents (Table 7 direction).

Anticipation / safety

Predictor residual as surprise; STA-style next-active object.

Sim-light WM pretrain

Household robotics without pixel video WM waste.

Education slogan

Representations should be predictive AND locally grounded.

Affordance probes

Frozen 2.1 vs SigLIP/DINOv3 on contact/depth frames first.

Anti-patterns — don't do these

Visionary discipline

✗ Drop-in SigLIP replacement without frozen-probe evidence (language cost).

✗ Cite Ego4D 7.71 or Franka +20% as if that moves BEHAVIOR q-score.

✗ Invent hybrid JEPA+FM as an existing demonstrated system.

✗ Plan Discord bots / Drive sync demos (explicitly out of scope).

✗ Jump to ViT-G course pretrain — prefer distill L/B + LoRA.

✗ Treat exclusive FM vs JEPA as settled — they're parallel bets.

Consistency triangle — where to bet

Ship today vs ablation vs WM parallel

SigLIP / DINO

Ship today
VLA towers

$\pi_{0.5}$ / OpenVLA
languageed policy

MAE / VideoMAE

FT-heavy ablation

pixels / tokens
not WM default

JEPA → 2.1 → 2-AC

Latent imagination

dense-state WM
goal-image CEM

Default language trunk stays SigLIP-class; V-JEPA 2.1 is the video/dense/WM ablation — Empiricist frozen probes first.

One-liner — tattoo this

Dense-state upgrade making latent planning spatially trustworthy

V-JEPA 2.1 is the dense-state upgrade to Meta's video world-model line — the piece that makes latent planning spatially trustworthy enough for grasping and fast navigation; BEHAVIOR hybrids should treat it as imagination backbone, not as a SigLIP replacement without measuring language cost.

Numbers tattoo — pin these

Reconciled with KANTA_PACK · shared across seats

0.307

NYUv2 RMSE linear
beats DINOv3-7B 0.309

22.2→47.9

ADE path T1
33.8→38.6→47.9

7.71

Ego4D STA mAP
ViT-G

40.8

EK100 Action R@5
ViT-G

77.7%

SSv2 attentive
72.8→62.5→72.1→77.7

60/70/80

Franka grasp n=10
matched→H=8 CEM

5.687

Tartan Drive ATE
10.6s vs NWM 103.2s

~10×

latent nav cheaper
than NWM SD-VAE

L_ctx

alone hurts globals
deep SS restores

33.8/62.5

T2 weighted $\lambda=0.5$
ADE / SSv2 trade-off

ViT-G 2B

VisionMix163M
+ cool-down 64@384

distill B/L

ViT-L $\approx$ G on
many probes (T11)

Visionary — close

Takeaways + exam bite

1. BEHAVIOR winners = $\pi_{0.5}$ +FM+SigLIP; JEPA = parallel latent imagination.
2. Hybrid JEPA+FM = PROPOSAL ONLY — not demonstrated.
3. Dense unlock = contact geometry / depth for household; follow-ups: AC on OXE/BEHAVIOR, goal-image CEM, dense aux.
4. NO Discord/Drive. One-liner: dense-state upgrade making latent planning spatially trustworthy.

EXAM BITE

Treat V-JEPA 2.1 as BEHAVIOR imagination backbone candidate; hybrid with FM is a proposal; never drop-in for SigLIP without probes.

Fig. 9 — latent imagination visual

10× nav planning — what 'spatially trustworthy' looks like

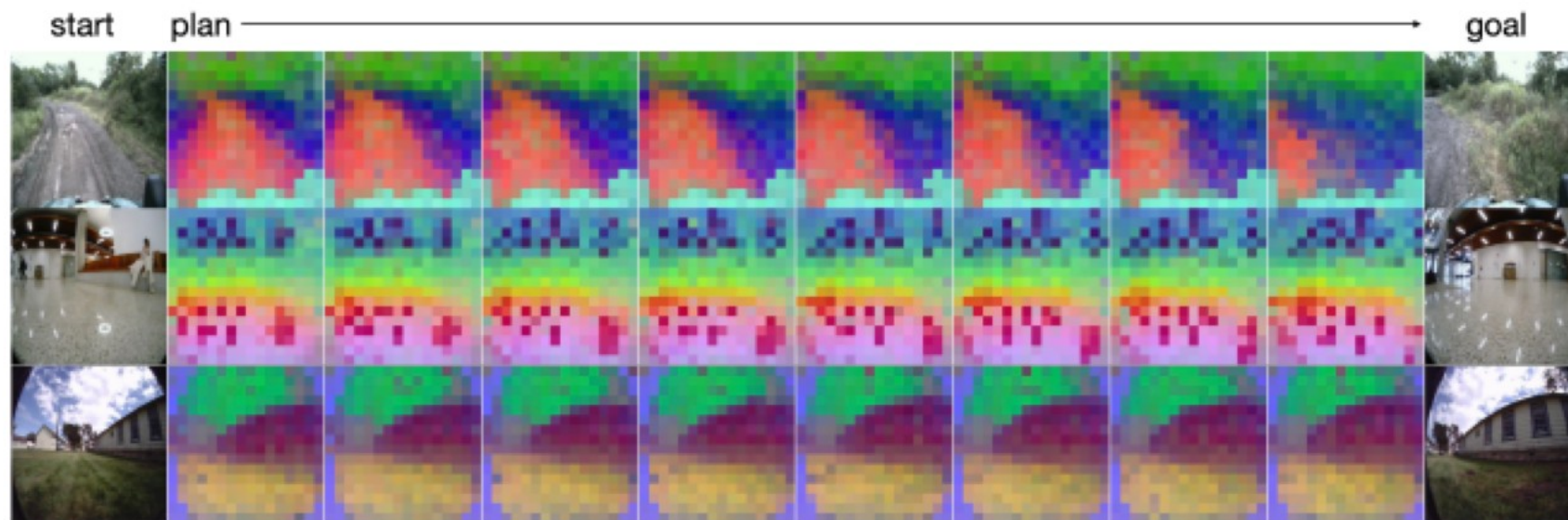


Figure 9 Planning Navigation Trajectories in Latent Space. V-JEPA 2.1 enables 10× faster and more accurate navigation planning compared to (Bar et al., 2025). We show PCA visualizations of 8 denoising steps of the planned latent trajectory between a start frame and a goal frame.

Table 7 Open Loop Navigation Planning. We finetune V-JEPA 2.1 on robot navigation datasets and compare the open-loop planning performance to NWM. Reporting the average planning time (seconds), normalized Average Trajectory Error (ATE) and Relative Trajectory Error (RTE). V-JEPA 2.1 representation enables 10× speed-up without sacrificing performance.

Cite Table 7; system confound OK to footnote; still the visionary picture of JEPA planning.